

WJEC GCE Unit 5

Condense Notes

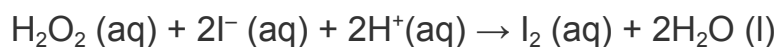
2026 Version

IGC HK Exam

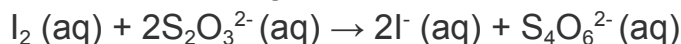


Kinetics – Iodine Clock

Reaction Equation



Sodium thiosulfate act as a delay



Overall Rate Equation

$$\text{Rate} = k [\text{S}_2\text{O}_8^{2-}] [\text{I}^-]$$

Questions

Use of H_2O – To allow the total volume to be same so volume of $\text{S}_2\text{O}_8^{2-}$ or I^- is proportional to their concentration + Fair Test

Colour change – Starch: Colourless to Blue-black

Calculate concentration of IV – $c_1 v_1 = c_2 v_2$ so eg I^- is IV $c_{\text{I}^-} v_{\text{I}^-} = c_{\text{IV}} v_{\text{Total}}$

More Objective Result – Colorimeter

More Reliable Result – Repeat experiment and take mean + Increasing range of concentrations

More Accurate Result – Higher resolution stopwatch

Rate Equation Calculation

Rate Increase Factor = [Zeroth Order]⁰ [First Order] [Second Order]²

$$\text{Ex: } x_4 = (x_2)^1 (x_2)^n n=1$$

Conclusion sentence: The reaction is nth order with respect to [conc].

Arrhenius Equation

$$k = Ae^{\frac{E_a}{RT}}$$

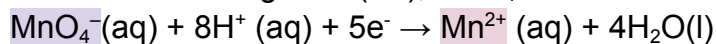
$$\ln k = \ln A - \frac{E_a}{RT} \times \frac{1}{T}$$

Mark Scheme

- Total volume (H_2O_2 / I^-) and H_2O must be same
- Table + Graph must include units

Feasibility of Redox

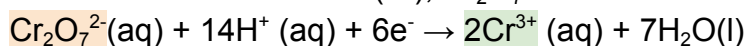
Manganate(VII), MnO_4^-



Purple $\rightarrow$ Pale Pink

MnO_2 – dark brown ppt

Dichromate(VI), $\text{Cr}_2\text{O}_7^{2-}$



Orange $\rightarrow$ Green

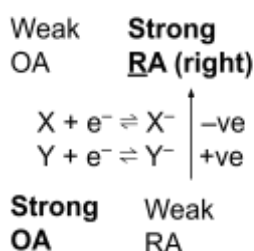
Salt Bridge

Function: Completes the circuit, allow a current to flow and keep equipotential

Substance: Filter paper soaked in saturated potassium nitrate solution

Standard Conditions
Temperature: 298K
Concentration: 1 mol dm⁻³ of HCl
Pressure: 1 atm

Platinum electrode: Inert, Conductor, Catalyst

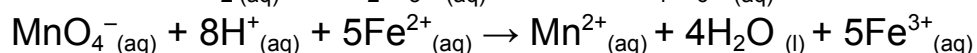
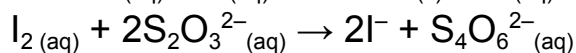
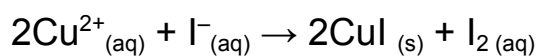


$$E^\theta_{\text{cell}} = E_{\text{reduction}} - E_{\text{oxidation}}$$

Half Equation



Memorise Equation



Compare Feasibility

Electrode reaction	$E^\ominus / \text{V}$
$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \rightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$	+0.96

Explain why copper does not react with most acids but does react with nitric acid

- Hydrogen electrode is less positive than the Copper electrode so H^+ cannot oxidise Cu to Cu^{2+}
- Nitrate electrode is more positive than copper electrode so NO_3^- can oxidise Cu to Cu^{2+}

Explain Feasibility

- Compare $E^\ominus_{\text{cell}}$ +ve / -ve
- As standard electrode potential for hydrogen peroxide is more positive than that for chromium/dichromate

- (c) Blue oxovanadium(IV) cations, VO^{2+} , can be oxidised to yellow oxovanadium(V) ions, VO_2^+ , by bromate(V) ions, BrO_3^- .

	Standard electrode potential, $E^\ominus / \text{V}$
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$	+1.00
$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.09
$\text{BrO}_3^- + 6\text{H}^+ + 6\text{e}^- \rightleftharpoons \text{Br}^- + 3\text{H}_2\text{O}$	+1.44

Use these standard electrode potentials to identify the main bromine-containing product formed when excess bromate(V) ions are added to a solution containing VO^{2+} ions. Give reasons for your answer. [3]

Br_2 (1) MUST GIVE REASON TO GAIN THIS MARK
 bromate is reduced to bromide / bromate is a strong enough oxidising agent to oxidise VO^{2+} as it has a more positive $E^\ominus$ value
 OR EMF = +0.44 V and this is positive so it is feasible (1)

bromate is a strong enough oxidising agent to oxidise the bromide formed in the first step to Br_2 as it has a more positive $E^\ominus$ value
 OR EMF = +0.35 V and this is positive so it is feasible (1)

Mention about more +ve / -ve SEP value

So it is a stronger OA / RA to oxidise / reduce the other cell

Chemical Test

Flame Test

Li ⁺	Red
Na ⁺	Yellow-orange
K ⁺	Lilic
Ca ²⁺	Brick red
Sr ²⁺	Red
Ba ²⁺	Apple green
Cu ²⁺	Blue-green

Other Test

$\text{SO}_4^{2-} \rightarrow \text{Ba}^{2+} \rightarrow \text{Heavy white ppt}$

$\text{CO}_3^{2-} \rightarrow \text{HNO}_3 \rightarrow \text{Effervescence} \rightarrow \text{Test CO}_2 \text{ gas by bubbling in lime water} \rightarrow \text{white ppt form}$

Halide $\rightarrow$ Acidified Silver Nitrate $\rightarrow$ AgCl (white ppt), AgBr (cream ppt), AgI (yellow ppt) $\rightarrow$ Dilute Ammonia $\rightarrow$ Concentrated Ammonia

	Colour	Dilute NH ₃	Conc NH ₃
AgCl	White	Dissolves	Dissolves
AgBr	Cream	Insoluble	Dissolves
AgI	Yellow	Insoluble	Insoluble

Group II

- Sulphate **decrease** solubility down the group ($\text{BaSO}_4 \text{ (s)}$)
- Group II Carbonate all insoluble
- Hydroxide **increase** solubility down the group ($\text{Mg(OH)}_2 \text{ (s)}$)

Group VII

Strong OA	Weak RA
Cl_2	Cl^-
Br_2	Br^-
I_2	I^-
Weak OA	Strong RA

Halogen	State at room temp	colour	colour of vapour	colour of aqueous solution
F	gas	pale yellow		very pale yellow
Cl	gas	pale green		very pale green
Br	liquid	red-brown	orange	orange
I	solid	grey	purple	brown

Solubility

Soluble Salt	Insoluble Salt
All Group 1 Salts	
All NH_4^+ Salts	
All NO_3^- Salts	
Most Halide	Halide salt of Pb^{2+} , Ag^+ , Cu^+
Most Sulphate	BaSO_4 PbSO_4 SrSO_4 Ag_2SO_4 CaSO_4 (sparingly soluble)
Group 1 & NH_4 Carbonates (mentioned above)	Most Carbonates
Group 1 & NH_4 Hydroxide $\text{Ba}(\text{OH})_2$ $\text{Sr}(\text{OH})_2$ Hydroxide	Most Hydroxides

Amphoteric:

Solution forms a precipitate in addition of hydroxide, which is then able to redissolve in excess hydroxide, the compound is amphoteric.

Chromium (Cr) – transitional metal

Zinc (Zn) – d block

Aluminum (Al) – p block

Lead (Pb) – p block

Precipitation of Pb^{2+} ions

$\text{Pb}^{2+} \rightarrow \text{Pb}(\text{OH})_2$ White ppt $\rightarrow \text{Pb}(\text{OH})_4^{2-}$ Redissolve colourless

$\text{PbCl}_2 \rightarrow$ White ppt

$\text{PbI}_2 \rightarrow$ Canary Yellow ppt

Manganese	Manganate (VII) MnO_4^-	dark purple
	Manganate (IV) MnO_2	brown
	Manganate (II) Mn^{2+}	very pale pink
Chromium	Dichromate (VI) $\text{Cr}_2\text{O}_7^{2-}$	orange
	Chromate (VI) CrO_4^{2-}	yellow
	Chromium (III) Cr^{3+}	green
Iron	Iron (II) Fe^{2+}	very pale green
	Iron (III) Fe^{3+}	rust orange
Cobalt	Cobalt (II) Co^{2+}	pink
	Cobalt (III)	blue
Copper	Copper (II) Cu^{2+}	sky blue
	Copper (I) Cu^+	colourless

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	Sky Blue
$[\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2]_{(s)}$	Pale Blue / 'Duck Egg Blue'
$[\text{Cu}(\text{H}_2\text{O})_2(\text{NH}_3)_4]^{2+}$	Royal Blue
$[\text{CuCl}_4]^{2-}$	Yellow
$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$	Pink
$[\text{CoCl}_4]^{2-}$	Blue
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	Pale Green
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	Rust Orange
$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	Pale Pink

	Some NaOH	Excess NaOH
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	grey-green ppt of $[\text{Cr}(\text{H}_2\text{O})_3(\text{OH})_3]$	ppt dissolves giving a deep green solution of $[\text{Cr}(\text{OH})_6]^{3-}$
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	dark green ppt of $[\text{Fe}(\text{H}_2\text{O})_4(\text{OH})_2]$	ppt does not redissolve red brown colour might be seen as Fe^{2+} might be oxidised to Fe^{3+}
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	red-brown ppt of $[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3]$	no further reaction
$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	pale blue (duck egg blue) ppt of $[\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2]$	no further reaction

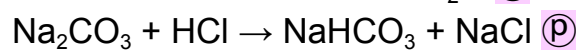
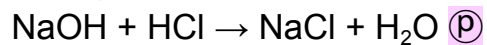
- $\text{Fe}(\text{OH})_2$ and $\text{Cr}(\text{OH})_3$ is much darker precipitate than the Fe^{2+} solution and Cr^{3+} solution
- Fe^{3+} is being oxidised from Fe^{2+} (rust iron)

Salt Preparation

Soluble Salt Crystal	Insoluble Salt
- <u>Excess</u> Insoluble + Acid Metal Metal Oxide Metal Hydroxide Metal Carbonate - Filter off the excess insoluble solution - Heat until <u>saturated</u> - Cool down for crystal to form - Collect by filtration - Wash deionized water, dry	- Dissolve the soluble salt in deionised water - Soluble salt for ion-exchange - Collect by filtration - Wash - Dry with filter paper

Acid and Base – Double Titration

Scenario 1 (NaOH + Na₂CO₃)



Scenario 2 (Na₂CO₃ + NaHCO₃)

